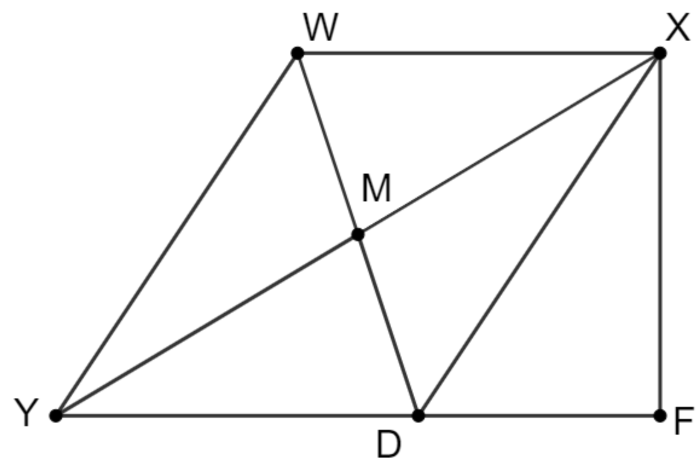
Geometry
Unit 5
Lesson 7 Homework

Name Answer Key_________
Date_____________________
Period____________________

Quadrilateral $WXFY$ is shown where M is the midpoint of $\overline{WD}$ and $\overline{WX} \parallel \overline{YF}$.



Given: M is the midpoint of $\overline{WD}$, $\overline{WX} \parallel \overline{YF}$.
Prove: $\overline{WX} \cong \overline{YD}$

Use the given information to answer the questions about the incomplete proof that is shown. Each empty box has the corresponding question number.

M is the midpoint of $\overline{WD}$

Given

Question 1

Question 2

$\overline{WX} \parallel \overline{YF}$

Given

Question 3

Question 4

Question 5

Question 6

Question 7

Question 8

Question 9

Question 10

1. What is the statement that belongs in the box?

   $\overline{WM} \cong \overline{MD}$

2. What is the reason for the statement?

   Definition of midpoint

3. What is the statement that belongs in the box?

   $\angle MXW \cong \angle MDY$

4. What is the reason for the statement?

   If two parallel lines are intersected by a transversal, the alternate interior angles are congruent

5. What is the statement that belongs in the box?

   $\angle WMX \cong \angle YMD$

6. What is the reason for the statement?

   Vertical angles are congruent

7. What is the statement that belongs in the box?

   $\triangle WMX \cong \triangle DMY$

8. What is the reason for the statement?

   ASA $\cong$

9. What is the statement that belongs in the box?

   $\overline{WX} \cong \overline{YD}$

10. What is the reason for the statement?

    Corresponding Parts of Congruent Triangles are Congruent